

Question_7:

Code:

```
1  /*
2  Question No: 7
3  A B-sequence is first strictly increasing and then strictly
4  decreasing.
5  The maximum occurs once. Every other value can occur at most
6  twice.
7  For every value, insert it only if the sequence can still be
8  a B-sequence.
9  After every operation print the size.
10 Finally print the B-sequence.
11 */
12 #include <iostream>
13 #include <map>
14 #include <vector>
15 using namespace std;
16
17 int main(){
18     int N,Q;
19     cin>>N;
20
21     map<int,int> cnt;
22     for(int i=0;i<N;i++){
23         int x;
24         cin>>x;
25         cnt[x]++;
26     }
27     cin>>Q;
28     while(Q--){
29         int x;
30         cin>>x;
31         int mx=cnt.empty()?0:cnt.rbegin()->first;
32         if(cnt[x]<2 && x!=mx)
33             cnt[x]++;
34         int sz=0;
35         for(auto p:cnt)
36             sz+=p.second;
37         cout<<sz<<"\n";
38     }
39     vector<int> v;
40     for(auto p:cnt)
41         v.push_back(p.first);
42     for(int x:v)
43         cout<<x<<" ";
44     for(int i=(int)v.size()-2;i>=0;i--)
45         if(cnt[v[i]]==2)
46             cout<<v[i]<<" ";
47 }
```

Output:

Test Case_1:

```
4
1 2 5 2
4
5
4
1
5
3
6
2
6
1 2 3 5 2 1
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
5
3 4 5 1 3
3
6
6
2
7
3
7
1 2 3 4 5 6 3
PS C:\Users\Prabin Yadav\
```